

## Research

# Design principles for e-learning platforms featuring higher-education students' enterprise systems end-user training

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## Abstract

Enterprise systems are complex information systems that are inevitable for companies' success. As enterprise systems are only successful when used continuously and efficiently by end-users, knowledge on how to use them has become an important skill for employees. Research and practice favor an early obtainment of these skills for employees. Thus, it is usual to train students in enterprise systems usage before they start their professional career. Even though in organizational settings e-learning based approaches gain momentum, adapting design instances of multi-purpose platforms like Udacity, Udemy or Coursera existing approaches directed at students are scarce. Nevertheless, a well-informed e-learning platform design can support the learning process. Therefore, we investigate the design of e-learning platforms featuring students' enterprise system end-user training. To address the lack of guidance on designing such e-learning platforms, we proposed four meta-requirements and ten design principles to increase students' learning success on e-learning platforms focusing on students' end-user training. To achieve this goal and to ground our results on empiricism, we collected data about e-learning-platform design from three data sources: (1) We reviewed the existing literature, (2) carried out a multi-case analysis, and (3) derived insights from implementing an e-learning platform for students' end-user training and thereby demonstrating the applicability of the design principles. Finally, we evaluated the implementation. Following the design principles, learning success should be increased by a flexible learning environment with easy access and use, propelled motivation, and fostered information exchange.

**Keywords** E-learning platform · Design principles · Enterprise systems · Enterprise systems education · Students' end-user training

## 1 Introduction

In times of accelerating digitalization, the implementation of enterprise systems, such as SAP S/4 HANA, Microsoft Dynamics, and Oracle ERP, is almost inevitable for companies to support their processes and remain competitive [1]. Enterprise systems are standardized software packages provided by specialized software vendors covering and supporting the majority of processes in an organization [2]. They integrate application systems across an organization and, thus, allow for smooth operation and proper management of the entire company. Besides these advantages, the failure rates of

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enterprise system implementations remain high [3]. Research and practice have identified user acceptance and users' skills for system operation as critical success factors of an enterprise system implementation [4–8].

End-user training has been identified by organizations potentially addressing these critical success factors [9–11]. However, we argue that it is already beneficial for higher-education students to experience enterprise systems end-user training before entering an employment relationship [12], as there is an increasing demand for skilled employees [13] and users need three to five years to become productive in enterprise systems use [14]. This can be rooted back to the high degree of complexity combined with the fact that enterprise systems are not claimed to be designed to collaborate with its users [15, 16] and the user experience, in general, is not perceived as positive [17, 18]. Nevertheless, future employees' skills and knowledge of operating the implemented enterprise systems are required for efficient system use [12, 19, 20]. This development is already reflected by the integration of courses on how to operate enterprise systems into the curricula of colleges and universities in the past years [13, 19]. The need for higher-education students' enterprise systems end-user training is also discussed in the literature [12, 21]. Usually, these trainings are offered as in-classroom trainings combined with hands-on sessions [12]. Nevertheless, this type of training has some severe downfalls, as for example the lack of flexibility, the lack of individualization of learning, and the constraints for participation [6, 12, 21, 22].

Over the past decades, a plethora of new training approaches aimed at students, such as e-learning [23], hybrid models combining classroom training and e-learning [24], and simulation trainings [25], have emerged to overcome these shortcomings. Despite more than two decades of research and existing new approaches, education and training on enterprise systems still present significant roadblocks for enterprise system implementation [26, 27]. Even though organizations intensively switched to e-learning formats for training purposes [12, 28, 29], and several software vendors already established e-learning platforms following the successful example of e-learning platforms such as Udemy and Udacity [e.g., 30, 31], these modern approaches are still scarce in higher education settings. Nevertheless, to create a holistic learning environment, it is immanent to implement a suitable e-learning platform [32, 33].

Higher-education students have differing requirements towards enterprise systems and e-learning platforms [34, 35], as their knowledge level in the context of enterprise systems is quite low and they need to learn a whole lot more than employees [36, 37]. Enterprise Systems are rather complex, and therefore, learning how to operate them, is a cognitive challenging task [12]. Learning complex materials involves many cognitive resources, which are limited and can be blocked by the learning process itself, the material that needs to be learned, and the way the learning material is presented towards the learner [38]<sup>1</sup>. In the case of enterprise systems learning, the learning material is quite complex and, therefore, is likely to block several cognitive resources, the usage of the e-learning platform should not be challenging [38–40]. Even though some effort has been made in explicating approaches to design e-learning-based enterprise systems courses for students to form a positive attitude towards enterprise systems [12, 19, 21, 41], research on how to design e-learning platforms for students' enterprise systems end-user training is still scarce. As enterprise systems end-user training tackles the problem of mis- or non-usage of enterprise systems, it is important, to design an e-learning platform to enable efficient learning as well as being appealing to the user. Against this backdrop, we strive to answer the following research question:

*How should e-learning platforms featuring higher-education students' training for enterprise systems be designed?*

For the derivation of design principles for e-learning platforms featuring students' end-user training that comfort the learning process, we followed an iterative approach informed by Moeller, Guggenberger, and Otto [42]. We base our derived design principles on the results of three analyses: (1) We incorporated a literature review to derive potential meta-requirements [43]. (2) We also conducted a multi-case analysis investigating successful e-learning platforms to extract potential design principles [44], and (3) included experience from the implementation of an end-user training platform, on which we demonstrated the applicability of the design principles. Finally, we evaluated the implemented design principles with the help of a questionnaire. Our ten design principles are a codified form of prescriptive design knowledge [45] for owners of e-learning platforms who want to implement efficient e-learning platforms for students' end-user training as a class of systems.

The remainder of this research paper proceeds as follows: We elaborate on the background of enterprise systems end-user training, e-learning platforms, and principles for the design of information systems in Sect. 2. In Sect. 3, we introduce our research design. In Sect. 4, we present our meta-requirements and design principles for students' e-learning platforms focused on enterprise systems end-user training. In Sect. 5, we demonstrate the application of the design

<sup>1</sup> This refers to the cognitive load theory [38, 40].

principles developed at an e-learning platform designed for students' enterprise systems training. In Sect. 6, we discuss our developed design principles followed by a brief summary of our results in Sect. 7.

## 2 Background

In today's educational landscape, enterprise systems have become integral components in academic curricula, facilitating administrative processes, communication, and learning experiences within academic institutions. These systems encompass a diverse array of software applications and online platforms designed to optimize operations and provide seamless experiences for users. Alongside enterprise systems, learning platforms play a pivotal role in modern education, offering versatile environments for collaboration, content delivery, and interactive learning experiences. These platforms, including learning management systems (LMS) and specialized educational tools, cater to diverse student needs and preferences. In this background section, we will first evaluate enterprise systems end-user training for students and afterwards introduce e-learning platforms, as well as their impact on learning.

### 2.1 Students' end-user training for enterprise systems

Enterprise systems are standardized and customizable information systems, that aim at integrating and optimizing a company's core business processes and transactions [46]. Among these processes are the procurement of goods, product sales, and decision-making within a company [2]. Thus, enterprise systems not only comprise an enterprise resource planning system, but also systems for customer relationship management, supply chain management, and business intelligence [2]. As these systems cover the support of nearly all important business processes within a company, they are quite complex to understand and difficult to operate by end-users [13, 46]. End-users are organizational units and individual employees, that consume and produce information by using enterprise systems [47, 48]. However, enterprise systems can only unfold their full potential if being used and handled efficiently [6]. To handle enterprise systems, end-users need to obtain technical, analytical, and business process skills as well as relevant corresponding knowledge [13, 19]. It is stated that for an average end-user to become efficiently productive in enterprise systems use, it takes three to five years [14]. Therefore, training for end-users on how to use these systems is advisable [6, 26, 27], and there is an increasing demand for skilled employees [13]. Therefore, it has become economically important to train employees as early as possible and foster the attainment of skills in enterprise systems use [19, 49]. Following this, it has become common to incorporate enterprise systems training into higher education curricula, to support students in attaining enterprise systems knowledge and skills [12, 13, 19]. Especially, current higher-education students and future graduates are considered as potential future users of enterprise systems [19]. Even though these students are considered to have an initial business understanding and are usually computer literate, many still struggle to be able to cope with enterprise systems [21, 25]. Nevertheless, as it is required for many entry-level job positions to have these skills, students should learn how to use enterprise systems. As research in this field is scarce, within this publication, we focus on students' end-user training for enterprise systems.

End-user training is defined as the process in which end-users acquire necessary information systems skills and knowledge, which enables them to operate certain information systems [22, 47]. Thus, the goal of end-user training for enterprise systems is to educate and motivate potential users and to provide them with the necessary skills to perform tasks within the enterprise systems [47, 50]. Because of the complex nature of enterprise systems compared to smaller information systems used within the company, training on how to use these enterprise systems is necessary, but difficult to design [12, 13]. Furthermore, we need to emphasize that end-users form a heterogeneous group characterized by different prior information systems knowledge levels, different predispositions to operate an information system, and non-uniform learning abilities and types [51]. Among the most cited disadvantageous aspects observed in traditional end-user training are a lack of flexibility and individuality [51, 52]. This also accounts for students.

We propose the application of e-learning formats as a reaction to the inflexibility and the commoditization of traditional end-user training [12, 13, 51]. Research in this field proposes new approaches, such as simulation games, to enable students to use enterprise systems [25]. However, even though there exist e-learning based approaches towards e-learning based enterprise systems end-user training [23] research in the field of designing efficient e-learning environments for teaching students to use enterprise systems is scarce. Especially, recommendations on how to design e-learning environments are lacking. Nevertheless, the fact that the traditional end-user trainings as well as e-learning formats are criticized for both being not sufficient makes a reconsideration of designing these environments necessary [6]. One

starting point to do so could be to design the end-user training platform with a stronger focus on the students' needs as end-users. As students usually have different knowledge level than employees, resulting from the lack of working experience and differing requirements to the design of e-learning platforms [34], we take these as focus group. Enterprise systems are only successful in organizations if the end-users use it efficiently [4].

As the e-learning environment is the first touching point of the students it can have an influence on the attitude towards the enterprise systems end-user training in general significantly [53–56]. Furthermore, as learning enterprise systems is a complex process [13], the platform through which the learning content is provided should not be too complicated to use. Therefore, the design of the e-learning platforms should be on focus, when keeping in mind to carry out an e-learning end-user training.

## 2.2 E-learning platforms

Due to recent developments resulting in the improvement of information technology, the interest in e-learning has significantly increased in recent years [57]. E-learning is a form of technology-mediated learning in which the learning material of a course is provided in an electronic manner [58]. Moreover, a virtual learning environment is provided which can be accessed via a web browser [59, 60]. Originally, e-learning was used for training information technology skills. However, meanwhile, it is also applied to business skills and other subjects in several domains [60, 61]. The provided course material can be distributed to the students in different formats (e.g., text and video). They can communicate with their fellow students within a course via several communication channels, such as chat rooms and forums [62]. The knowledge gain of students, who participate in an e-learning course, can be tested and evaluated [63]. Even though e-learning is a topic of interest in research and practice, no generally accepted definition has yet been established. Our understanding of e-learning is inspired by the European Commission. [64]. [64]. According to this definition, e-learning includes using multimedia technologies and the internet to improve the quality of learning by facilitating access to learning materials [64].

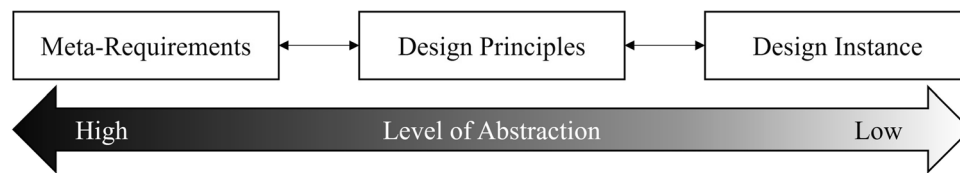
To introduce e-learning platforms, we draw on research on (digital) platforms [65, 66]. In general, platforms are either conceptualized as a technological foundation providing application services upon which third-party developers can implement platform extensions (i.e., innovation platform) [66–70], or as (virtual) environments matching different types of participants and enabling business transactions between them (i.e., transaction platform) [65, 67, 68]. Platforms combining both concepts are described as hybrid platforms [71, 72]. E-learning platforms are considered transaction platforms enabling the interaction between learner and content providers [73]. Following archetypal roles in platform business models introduced by Eisenmann, Parker, Van Alstyne, et al., lecturers act as supply-side participants providing knowledge in a purified form customized for a particular audience (i.e., demand side) [74]. A platform owner operates the platform (e.g., Coursera, Udacity).

E-learning platforms provide a cost-effective solution for corporations and individuals to distribute knowledge [58]. Especially, different age groups in different areas can be reached with the help of an e-learning platform [75]. The activities supported by an e-learning platform are creation, organization, delivery, communication, collaboration, and assessment [73]. Within the platform, the lecturer is enabled to create content and update existing material [76]. Further, this content is organized as a module or a course and delivered in those, so that students can access them [73]. To foster the collaboration among students several communication channels can be implemented enabling synchronous and asynchronous communication [29]. While the first form of communication refers to real-time communications between participants such as chat rooms or videoconferencing, the latter form includes a time delay between messages (e.g., forums and emails) [58]. Finally, assessments in the form of tests and quizzes can provide a way to verify the learning progress of students [76].

## 2.3 E-learning platform's impact on learning

The success of the implementation of enterprise systems depends on the level of usage [77]. Following the implications of the technology acceptance model, the system adoption is influenced by the perceived usability and perceived ease of use of a system [1, 78, 79]. Perceived ease of use and perceived usability<sup>2</sup> can be influenced by the user's satisfaction

<sup>2</sup> Usability can be defined as the extent to which a user can use a product in a specific context to achieve certain goals efficiently, effectively and to the users satisfaction [176].



**Fig. 1** Relation of meta-requirements, design principles and design instances. These differ based on their level of abstraction. The level of abstraction in meta-requirements is high. It decreases over design principles and is the lowest at design instances

with the system and its attitude towards the system. Furthermore, it has been proven, that a positive attitude towards a system leads to a higher motivation to use this system, and therefore is considered an impacting factor for system adoption [14, 27, 80]. Additionally, recent research has identified that training has an impact on the perceived ease of use and the perceived usability and, therefore, on the satisfaction with the system [81, 82]. Thus, to achieve continuous and efficient use, the attitude of the end-user needs to be focused [27]. The attitude of the students should be positive towards the enterprise systems in order to enable motivation to learn it [27, 49]. The attitude towards enterprise systems can be formed by the enterprise systems itself, the training for the enterprise systems and—nonetheless—the way in which the training material is provided towards the students [20, 83]. Therefore, to create a holistic user experience for the user, it is important to emphasize an informed training design. One aspect that needs to be considered in this regard is the way of presenting the information to the user. This can be done as classroom training or by any digital means [12]. In context of this publication we focus on provisioning the training material via digital means and focus on the design of e-learning platforms.

### 3 Methodology

In this section we will elaborate on the methodology on how we derived the meta-requirements and design principles for e-learning platforms featuring higher-education students' enterprise systems end-user training. Hence, we define design principles in information systems research in a first step. Afterwards, we elaborate on our approach for the development of design principles incorporating supportive and reflective design iterations.

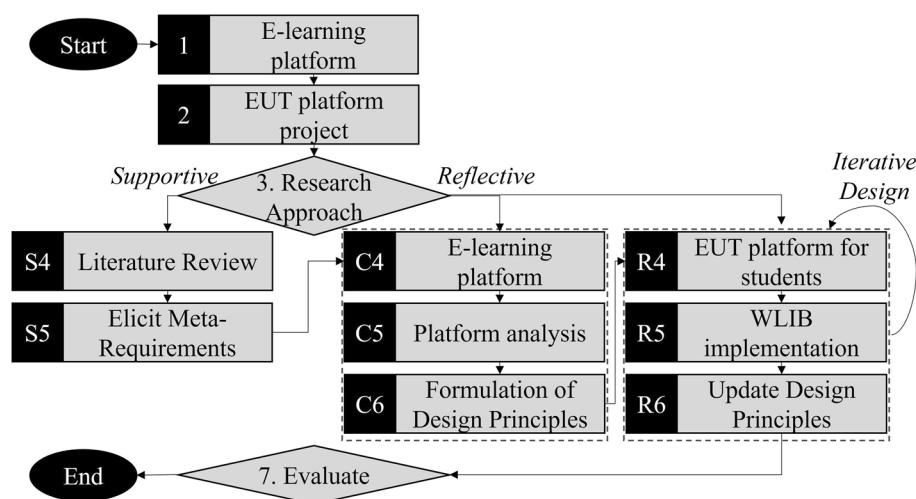
#### 3.1 Principles for information systems design

The implementation of information systems, such as e-learning platforms, can be guided by abstract design principles [84, 85]. Design principles are a form of prescriptive design knowledge specifying the design of an artifact [42, 45, 86]. Design principles "define the structure, organization, and functioning of the design product or design method" [87]. Moeller, Guggenberger, and Otto [42] defined design principles as "fundamental propositions that aid designers in achieving a successful transfer of requirements to design" [42, 86] and capture "knowledge about instances of a class of artifacts" [88]. This research paper aims to provide guidance for the design of e-learning platforms supporting students' enterprise systems end-user training. The concept of design principles has progressively entered design science research within information systems research to "supplement constructs, methods, models, and instantiations" [45, 89].

Meta-requirements can guide the derivation of design principles, which must be met by the design principles and the resulting information system, which will result as instances of the design [90–92]. As design principles codify design knowledge applicable for a class of systems [86, 88], they abstract from concrete design instances observable in implemented information systems. The implementation of design principles in concrete design instances, resembling (parts of) the actual information system, can be supported by design features as intermediary solution [90]. The types of prescriptive knowledge differ in the degree of abstraction from the actual design instances implemented (Fig. 1). We focus our design knowledge derivation on abstract design principles with meta-requirements.

In the following we explain how we derived these design principles and meta-requirements.

**Fig. 2** Research approach proposed by Möller et al. [42], that we followed to develop scientifically grounded and empirically validated design principles following supportive and reflective design iterations



### 3.2 Procedure for design principle development

The goal of this research paper is to propose a set of design principles for the implementation of e-learning platforms providing knowledge for students as enterprise systems users. Therefore, we followed the approach introduced by Möller et al. [42], to develop scientifically grounded and empirically validated design principles following supportive and reflective design iterations. We applied the template proposed by Chandra et al. [86] for a structured formulation of our design principles. Since design principles are nascent theories [93], we devised an approach to generate design principles from three data sources suitable to generate theory [42, 88, 94]. We conducted a literature-based supportive design iteration to identify meta-requirements for the design of e-learning platforms followed by two reflective iterations [44, 88]. Our research approach is depicted in Fig. 2. In the following we will explain the research approach more in detail. (1) The objective of our research is to create a successful e-learning platform. (2) The research context is specified in the previous section. We introduced enterprise systems end-user trainings especially focused on students. Furthermore, we introduced the issue of missing guidance for creating specified end-user training e-learning platforms. (3) As research approach we followed both a supportive and reflective perspective to derive our design principles. The supportive perspective aims to assist the creation of design principles ex-ante, while out of the reflective perspective the design principles are derived during the design iterations [42]. (S4) Following the supportive perspective we used a literature review to (S5) elicit our meta-requirements. The reflective iterations are informed by an in-depth analysis of nine successful e-learning platforms and the development of an e-learning platform featuring end-user training for students [44, 88]. The meta-requirements that were elicited in the supportive iteration defined the scope for the reflective approach. (C4) In a first step we identified 9 successful e-learning platforms, to perform a (C5) platform analysis on them. (C6) Resulting from this analysis we were able to formulate design principles. (R4) Based on these derived design principles we were able to develop iteratively an e-learning-based end-user training platform for students. (R5) This resulted in the implementation of our platform we learn in bits (WLIB). (R6) During development of WLIB we were able to update and adapt the design principles according to the experiences we made. (7) Finally, we were able to evaluate the implemented design principles, to test for whether they apply.

### 3.3 Supportive derivation of meta-requirements

To inform the design of an information system, requirements should be specified. Nevertheless, each user has different requirements for a system. To make these requirements manageable, meta-requirements need to be formed [95]. Meta-requirements describe classes of goals [92, 96]. To identify the meta-requirements we followed the supportive approach [42] and extracted meta-requirements from the current body of literature [92]. Therefore, we conducted a systematic literature analysis following [43]. We started our literature review with a (1) journal search focusing on leading information systems journals (i.e., basket of eight) complemented by proceedings from major information



systems conferences (e.g., International Conference on Information Systems). For our (2) database search, we used Web of Knowledge, EBSCO ScienceDirect, SCOPUS, Taylor and Francis Online, Springer Link, AIS eLibrary, and Google Scholar. We assessed the outlets by their titles, abstracts, and keywords. We performed our iterative (3) keyword search starting with the terms "e-learning platform" and "design". Moreover, we extended our search by adding the terms "learning platform" and "virtual learning environment". Following the approach proposed by Bandara et al. [97], we applied quality criteria to ensure a high rigor of identified papers. We especially searched for articles examining requirements for and design of e-learning platforms. In total, the literature search resulted in 13 relevant articles. Finally, we conducted a one-way (4) backward and forward search [43]. We identified another nine articles and ended with 22 relevant publications in total. We classified an article as relevant if it addresses the topic of designing e-learning platforms in general. The articles identified have been published between 2007 and 2022. The majority (13) of the identified articles deal with the topic of e-learning design; four describe how to design e-learning-based (end-user) training, three describe the design of end-user training in general, and only one identified article investigated design principles for e-learning. Table 1 in Appendix A provides an overview on the identified articles and their contribution to the derived meta-requirements.

By examining the identified articles in detail, we could examine features, that should be fulfilled by an e-learning platform to enhance students' enterprise systems knowledge. Clustering these features, we were able to derive meta-requirements for end-user training e-learning platform design.

### 3.4 Reflective derivation of design-principles

After the derivation of meta-requirements, we performed two reflective iterations to derive design principles to fulfill the identified meta-requirements [92]. We abstracted from concrete design instantiations of successful e-learning platforms, codified them into design principles, and augmented these findings with our own experiences from the development and establishment of an e-learning platform for students' end-user training. The inclusion of a multi-case study of successful e-learning platforms increases the "likelihood of generating novel theory" [98]. The subsequent design of an e-learning platform for end-user training allowed us to further shape the design principles [42]. To ground our research in empiricism, we report on a multi-case study of nine successful e-learning platforms for common design features [98]. The multi-case analysis enabled us to draw on design knowledge collected from a series of cases that might differ in environmental aspects but share a common phenomenon [44]. The selection of multiple cases allowed the identification of cross-case patterns and a comparative analysis of cases [99]. Our selected cases share the phenomenon of successful transaction platforms in the context of e-learning matching lecturers as well as content providers and learners. Moreover, the e-learning platforms selected provide end-user training for enterprise systems by software providers, among others Microsoft, Oracle, SAP, and Salesforce [100].

We analyzed two types of e-learning platforms: (a) multi-purpose e-learning platforms (Udemy, Edx, Coursera, Udacity, Iversity, Skillshare, Futurelearn) and (b) platforms focusing on end-user training for enterprise systems (OpenSAP, Trailhead by Salesforce). We provided an overview of our cases analyzed in Appendix B. We leveraged a plethora of publicly available data sources to collect information on each case. We analyzed only official information on each case and also screened secondary information portals, such as newspapers and repositories [94]. Exemplary sources analyzed for each case are depicted in Appendix B. The analysis of multiple cases required the abstraction of generally applicable design knowledge for a class of artifacts (i.e., e-learning platforms for end-user training) [86]. As guidance for the analysis, we used the features, derived from the literature, as well as the meta-requirements to generate a scheme, to make the results from analysis comparable. We present selected design instances analyzed from the e-learning platforms to further illustrate the developed design principles.

Based on the derived design knowledge, we re-implemented an e-learning platform for the enterprise systems end-user training focused on students. Taking into account the derived design principles, we created a new platform called *We Learn in Bits* (WLIB) [101]. To set up this platform a Moodle environment has been chosen as the technical basis. As Moodle is often used in educational purposes, and most students are familiar with it, it has been chosen as a technical basis, to make handling easy for students. "Moodle" is an acronym for "Modular Object-Oriented Dynamic Learning Environment" and describes an open-source course management system and learning platform [102]. Moodle currently has about 166,000 sites in 242 countries and a base of about 350,000,000 users. Germany is among the top three countries of use with 10,139 registered Moodle pages, followed by the United States (12,981 pages) and Spain (12,390 pages) [103].

Two researchers from the team of authors have been directly involved in developing the e-learning platform by contributing to the requirements analysis, implementation, operation, and evaluation. Hence, we could further enrich

our set of design principles based on the experiences gathered in their implementation and could demonstrate their implementation in actual design instances [42]. We had access to many project materials, such as deliverables, source code, project documentation, meeting notes, and students' inquiries. We retrospectively analyzed the implementation steps and the consequences of certain measures we made. The courses on the WLIB platform are designed in a multimedia micro learning format and aim at providing students with skills to operate enterprise systems, which they might need in their future work.

We started in 2019 with a default design and recognized that about 70% of students enrolled at that time, claimed that the design of the e-learning platform was not appealing. Furthermore, many students showed a lack of motivation to work in this course, which was demonstrated by drop-out rates about 75%. To encounter this we iteratively adapted the design of the e-learning platform, based on the insights gained from the case study analysis.<sup>3</sup>

#### 4 Meta-requirements and design principles for enterprise systems end-user training e-learning platforms

In the following chapter, we present the four meta-requirements and ten design principles derived. Students should be able to learn spatially and in a timely manner. Furthermore, there is the need to learn continuously [75]. Therefore, the e-learning platform should enable efficient and flexible learning (MR1). An aspect that contributes to this, is that the students should be able to access the e-learning platform without any complications and burdens, such as complicated and long registration processes [29]. Furthermore, the students need the possibility to access the platform permanently [104]. As enterprise systems are complex, learning how to use them requires a lot of mental effort [105]. The students need to be able to use the platform easily to be able to focus on learning the enterprise system. Therefore, the e-learning platform should be easy to use (MR2). Since enterprise systems are such complex systems, learning how to use them is not always quick and easy [13]. Additionally, companies often fail to explain to end-users the benefits of using the system [11]. To keep students on track, they need to be motivated continuously [104]. Therefore, the e-learning platform should enable students' continuous motivation (MR3). Learning is a process and especially when learning to use a system there might occur problems, for which the students need to be provided with fast feedback and maybe also help in order for them to get not frustrated with system use. Therefore, the e-learning platform should enable the exchange between all platform participants (MR4) [104]. We developed ten design principles addressing the formulated meta-requirements: Accessibility (DP1), permanent provision of Learning Content (DP2), Interactive tasks (DP3), clear structure and simple navigation (DP4), only course related aspects (DP5), multimedia (DP6), didactical elements (DP7), track of course progress (DP8), motivational elements (DP9), and communication (DP10).

The mapping diagram in Fig. 3 depicts the fulfillment of the identified meta-requirements by the developed design principles, with links between them, and provides each design principle with a short textual description [42]. As the meta-requirements are inter-related, there are some design principles that fulfill more than one meta-requirement (Fig. 3). In the following the design principles will be explained in further detail.

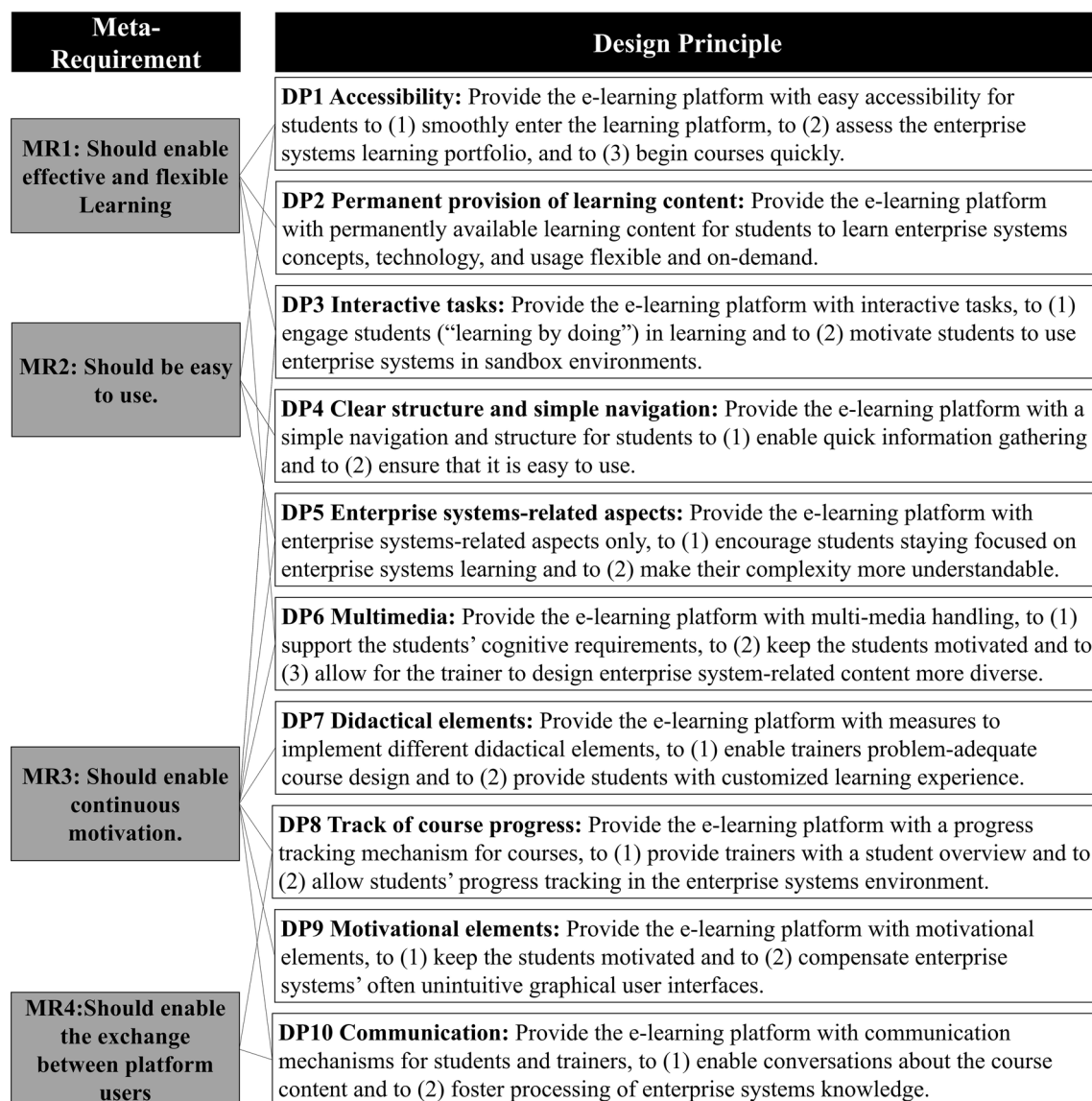
The main goal of this research endeavor is to provide prescriptive design knowledge for the implementation of an e-learning platform for students' enterprise systems end-user trainings. These trainings should improve the students' self-efficacy in enterprise systems usage [105]. To be effective, e-learning environments should be designed to accommodate the nature of the human mind and take into account the inherent limitations of perceptual and cognitive systems [106].

**DP1:** *Provide the e-learning platform with easy accessibility for students to (1) smoothly enter the learning platform, to (2) assess the enterprise systems learning portfolio, and to (3) begin courses quickly.*

Learning is defined as the process of acquiring knowledge [107]. It can affect and change an individual's behaviors, attitudes, skills, habits, and feelings [108]. Attitude within this context is defined as a person's favorable or unfavorable evaluation towards an object [109, 110]. It is hypothesized that a student's attitude can be affected by learning processes [111]. Thus, students will have a more positive attitude towards an enterprise system, if they believe, that the usage will have positive consequences [105, 112]. The provision of adequate training can tone this. Within this research paper, we consider the students' attitude towards enterprise systems as well as the students' attitude towards the e-learning

<sup>3</sup> See Appendix: for a detailed description of the iterative design.





**Fig. 3** Mapping diagram of design principles for e-learning platforms is depicted, illustrating which of the ten design principles match the four meta-requirements and allocating these

platform, as the perception of the learning material (in this case the e-learning platform) might determine the students’ attitude [14, 27, 80]. Therefore, students’ skills in using the enterprise system are considered to be affected by their perception of the e-learning platform [105]. Therefore, students should be able to easily and independently find and access the e-learning platform when they need it to allow for self-directed learning [58]. An e-learning platform can provide its users with one single access point with an overview of all enrolled learning units [113, 114]. This overview can also be used to start courses quickly. From here, a user can continue his enrolled courses or propose new courses to the user [115].

**DP2:** *Provide the e-learning platform with permanently available learning content for students to learn enterprise systems concepts, technology, and usage flexible and on-demand.*

One of the benefits of e-learning is that learning can be designed in a flexible manner [12]. Students can access the learning content independently of their time and location, which means that the learning content is provided permanently [75]. Usually, traditional enterprise systems end-user training takes place at a certain point in time, during or shortly after the implementation of the enterprise system [12]. Not all students who receive end-user training use the system daily, so the knowledge learned during the training might fade away over time when not used [6]. Students’ end-user training is also often delivered as in-class training distributed over the course of a semester [12, 13, 21].

Enabling a more flexible learning environment, students may want to start training at a more suitable time and may want to recapitulate the training or at least parts of it [116]. For the transfer of the learned content, learning at the point of the actual use would also be helpful [75]. Learning content can be made available permanently by archiving the course after its completion so that users can still access the course materials whenever they need to [117, 118].

**DP3:** *Provide the e-learning platform with interactive tasks, to (1) engage students ("learning by doing") in learning, and to (2) motivate students to use enterprise systems in sandbox environments.*

Students' engagement to learn is important within complex learning contexts, such as enterprise systems [119]. The trainer's engagement and motivation cannot be fostered in the same way in e-learning environments as in traditional training [41, 116, 120]. Therefore, there is a need to engage and motivate students via interactive tasks [19, 41, 120]. To create those interactive tasks sandbox systems are used, to foster student's understanding. Thus, the students can be kept motivated by learning-by-doing. Thereby, the students do not need to fear of deleting or disturbing business-relevant data. Therefore, students are engaged to learn by trial-and-error [21, 25]. The students can be kept motivated by the integration of interactive elements and promoted platform features, such as questionnaires, polls, and case studies, in the context of e-learning [29, 119, 121, 122]. E-learning platforms supporting massive open online courses should also provide a discussion forum to allow interactions among students [123, 124].

**DP4:** *Provide the e-learning platform with a simple navigation and structure for students to (1) enable quick information gathering and to (2) ensure that it is easy to use.*

The e-learning platform for students' enterprise systems end-user training should be easy to use to foster the acceptance and usage of enterprise systems [125]. Simple navigation and a clear structure of the e-learning platform are inevitable for the platform's ease of use [105]. Moreover, the e-learning platform should provide learning content especially tailored for students on how to use an enterprise systems properly [126]. As enterprise systems are quite complex to use, the usage of the e-learning platform should not be a cognitive challenging task [104]. The search for a course can be eased by providing a comprehensive search function with selected filters [31]. Furthermore, a clear structure can be propoled by providing categories for the subjects of the learning units [127] and for different kinds of courses the platform provides [122].

**DP5:** *Provide the e-learning platform with enterprise systems-related aspects only, to (1) encourage students staying focused on enterprise systems learning, and to (2) make their complexity more understandable.*

There are only a few platforms offering end-user training for complex enterprise systems [30, 31]. Platforms addressing students' enterprise systems end-user training are even more scarce. There are only a few approaches towards the design of such a platform [126]. Most of the other multi-purpose platforms also provide other courses in different areas. This might lead to the fact that the students get distracted from the actual content of focus because they might be interested in other topics as well [128]. Nevertheless, in order to design the platforms not to be overwhelming and help students focus on the topic of enterprise systems learning, the design should be easy and do not include unnecessary information [105, 126].

**DP6:** *Provide the e-learning platform with multi-media handling, to (1) support the students' cognitive requirements, to (2) keep the students motivated, and to (3) allow for the trainer to design enterprise systems-related content more diverse.*

Learning should support the cognitive requirements of the human brain [106]. Learning implies that the information obtained need to be stored in the long-term memory of students and be re-callable [38, 40, 129–131]. Even though the long-term memory is considered to have a vast capacity [132–135], the working memory represents a bottle neck for learning [38, 40, 106, 135, 136]. Among other factors, the complexity of the learning content is considered to have an impact on the capacity utilization of working memory, and might cause to overstrain this [38, 40]. As already stated are enterprise systems considered to be more complex compared to traditional information systems. To overcome this, it is suggested to follow the implication that information proposed in different media is processed in different parts of the working memory [129]. Multimedia is supportive for processing complex information and enables effective learning [5]. Therefore, multimedia should be used in students' enterprise systems end-user training to support the cognitive processes. Furthermore, students might feel more motivated as the change of media is more of a variety for them [137]. Videos, texts, audio or quizzes can help to accomplish such variety of multimedia [138, 139].

**DP7:** *Provide the e-learning platform with measures to implement different didactical elements, to (1) enable trainers problem-adequate course design and to (2) provide students with customized learning experiences.*

There exist different didactical concepts that can be used for the design of students' end-user training. Besides traditional classroom concepts, blended learning formats, simulation games, and case study approaches are often used in this context [140]. Focusing on e-learning formats, computer-based trainings, web-based trainings, game-based learning, serious games, and microlearning are often referenced [12]. The micro-learning approach might be helpful to disentangle the complexity of enterprise systems end-user training in general and for students' training in particular [141]. The provision of learning content in small units contributes to an effective learning process because the human brain can only process seven plus/minus two chunks of information [142]. A possibility to foster human learning is to provide small units of videos that can be watched whenever a student wants to [143]. Furthermore, trainers are free to design their courses addressing problem-adequate learning with different didactical elements, such as videos, projects, or quizzes [114].

**DP8:** *Provide the e-learning platform with a progress tracking mechanism for courses to (1) provide trainers with a student overview and to (2) allow students' progress tracking in the enterprise systems environment.*

In complex learning settings that take a lot of time to learn, like with enterprise systems end user trainings [120], it can be helpful for students to track their course progress in order to see how many they have already achieved [144, 145]. Generally, the motivation needs to be kept up during the learning process [120, 126, 146]. Furthermore, the trainer might want to know the different levels of progress within the overall course to be able to provide certain hints and to emphasize some topics [147, 148]. Both approaches might help to foster student's motivation during the learning process [148]. This is why on many platforms, the students can track their progress in each course they are enrolled in within their individual accounts [114] and through an overview provided in the enrolled course itself [149]. For the trainers on the e-learning platform, a similar overview about the number of enrolled students and the number of units consumed within an individual course helps to provide an overview of the course and the progress of students [127].

**DP9:** *Provide the e-learning platform with motivational elements, to (1) keep the students motivated, and to (2) compensate enterprise systems' little motivating user interfaces.*

As motivation is an important success factor for students' learning [150], the e-learning platform should feature motivational elements [120, 126]. Especially in enterprise systems context, where user interfaces are considered to not strongly collaborate with its users and to be low in user experience [17, 18], this is of importance [5, 25, 27, 151]. Gamification elements and certifications can be used by trainers, to foster the interest and the engagement of students in enterprise systems and to increase motivation [146]. This includes challenging the students to achieve certain goals (in a certain time), or to include interactive parts, so that students feel involved and engaged [152]. Goals can take the form of different achievements such as badges, which can resemble rewards for the student, so that the learning experience feels more like a game than like learning [153]. Those rewards can be provided as certifications for the successful completion of courses which students can download [149, 154].

**DP10:** *Provide the e-learning platform with communication mechanisms for students and trainers, to (1) enable conversations about the course content and to (2) foster processing of enterprise systems knowledge.*

The social interaction is sometimes lacking on e-learning platforms and in e-learning courses. Although the community feeling and the personal interaction with the lecturer are not perceived as important in context of enterprise systems training [24], the direct feedback in the case of problems is important as motivational means [111, 116, 129]. The exchange between the participants on the e-learning platform (students and trainers) might lead to an improved processing of the information learned, as it needs to be verbalized [58]. Furthermore, communication among students and between trainers and students might be beneficial for carrying out tasks in the enterprise systems [129]. Therefore, trainers and students can communicate with each other in a forum open to everyone enrolled to the course for questions, ideas and views on specific topics [155]. Another aspect is, that the students' end-user training should be especially designed for students as end-users, as they likely have different knowledge levels compared to professionals [25]. So to get hints on how to improve courses, a feedback option provided by the e-learning platform would be beneficial, which can be achieved by for example giving a rating in form of stars or a short comment about the course in general [156]. Also, trainers can announce their improvements to the students to acknowledge their

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**Exam: ES and SAP**

Quiz   Question bank

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**Question 2**

Not yet answered

Marked out of 1.00

Flag question

v6 (latest)

How can SAP-Fiori-Applications be classified?

Choose the correct answers.

- ☐ a. Training related
- ☒ b. Transactional
- ☐ c. Infoblatt (factsheet)
- ☐ d. Analytical

Previous page   Next page

**Quiz navigation**

1 2 3 4 5

Finish attempt ...

?

**Fig. 4** Integration of interactive elements on our platform. This is a showcase of an interactive quiz we implemented in each section, to foster students' engagement with the training material

feedback in either a dedicated forum or in the aforementioned open forum [118]. With the provision of a frequently asked question section, most problems can be dealt with [157]. Moreover, an integrated chatbot can help speed up to find a solution for a problem [158]. Lastly, a community forum for the whole e-learning platform can be used to solve problems or answer questions [159].

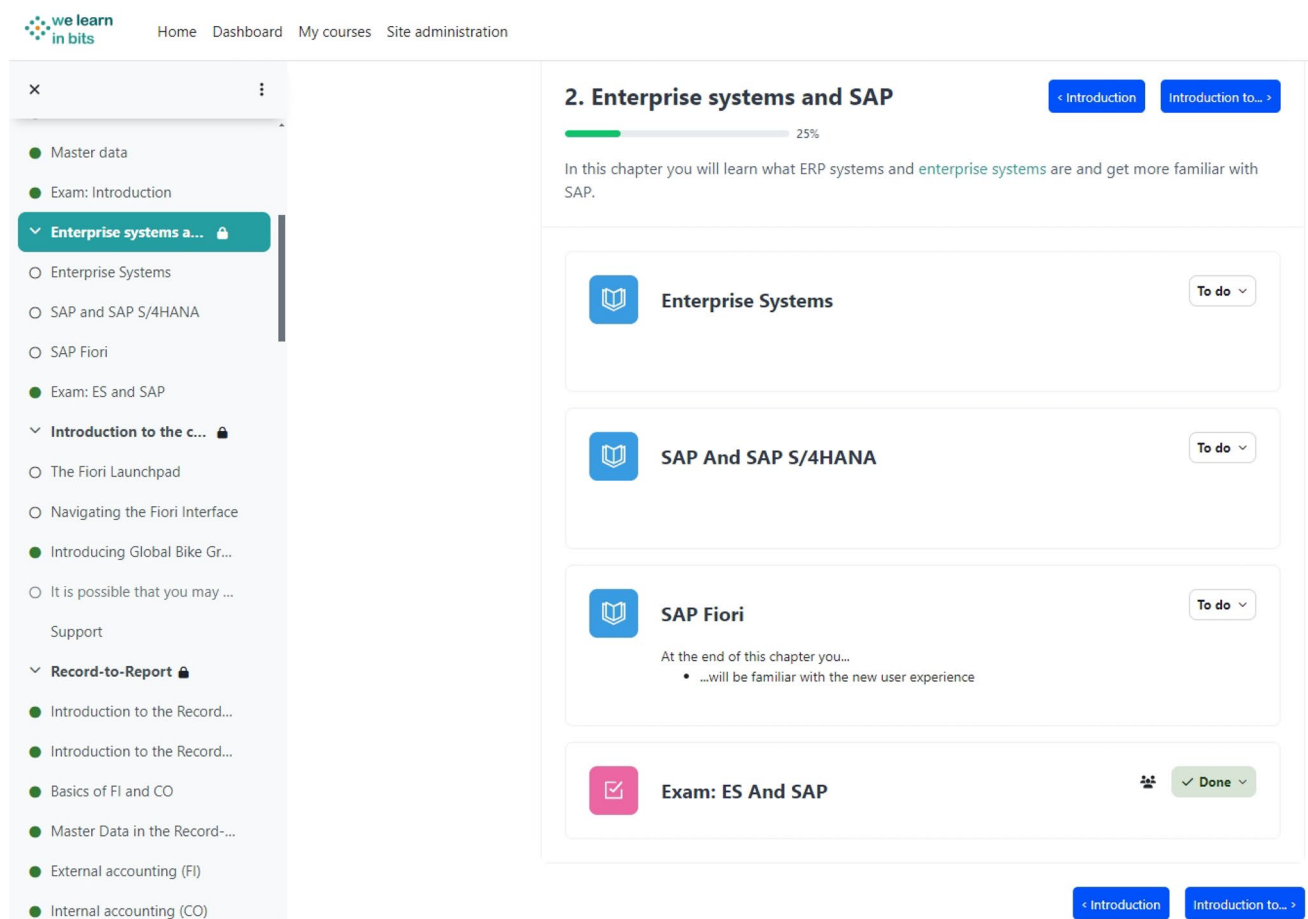
## 5 Demonstration

In 2021, we started the re-implementation of our e-learning platform for higher-education students. We made use of the design knowledge acquired to restructure the platform, adapt the didactic and pedagogical concept, and update the entire user interface.

During the development of our platform WLIB it has been proven, that students quickly gain a negative attitude when accessing the platform is too complicated, and handling is not easy. As a first measure, we improved the links between our website and the e-learning platform making the platform accessible via the website and findable via simple web searches. For the WLIB platform, we implemented a single-sign-on-mechanism that simplifies the log-in for the e-learning platform and the adjacent webshop. This log-in mechanism harmonized accounts and, thus, reduced student's disturbance and distraction. The enhanced account management is also used to send the students links for entering the e-learning platform when starting the course.

Addressing design principle 2, we restructured the courses to support students' individual schedules with learning pauses and flexible continuation. Hence, students can benefit from the course contents whenever it suits them best. We disengaged from the initial semester-wise enrollment phases and enabled a flexible enrollment all throughout the year for the students. This new enrollment allowed the students to individually plan the course conduct and arrange it with other activities. Moreover, these two measures allowed us to continuously publish new courses and provide additional learning materials.

For the relaunch of WLIB, we integrated interactive tasks in our courses so that students feel more engaged and motivated. We created interactive quizzes that engage the participants to reflect on the theoretical contents (Fig. 4). Furthermore, the e-learning platform is designed to support practical exercises and case studies to foster the enterprise systems handling knowledge of students. For the implementation of the interactive elements, we make use of the standard elements (e.g., assignments, quizzes, wikis, chats, h5p, forum) provided by the Moodle platform used as a basis.



**Fig. 5** The navigation on our WLIB platform. We styled the sections in each chapter as tiles, so that the structure of the navigation bar corresponds to the structure within the chapter. This way students can easily navigate on the platform

We also augmented this set of interactive elements with selected third-party modules (e.g., virtual meetings, SCORM packages, video animations). Moreover, we updated our guidelines for course creation and made the use of interactive elements a mandatory ingredient of new courses.

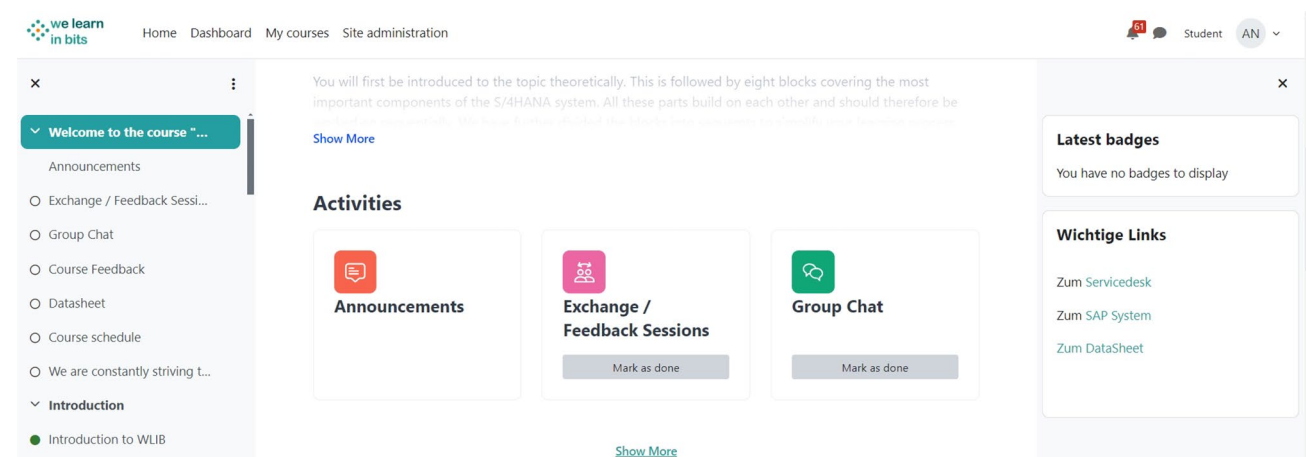
Concerning design principle 4, we observed a too complicated menu structure for WLIB's predecessor. Reportedly, this navigation structure led to a high level of frustration among the students. Therefore, we redesigned the navigation of the e-learning platform and offered the overview of courses displayed as tiles to get a clear overview of the courses that the participants are enrolled in (Fig. 5). We also reflected the simplified navigation in the structure of the courses. The chapters of each course are displayed as tiles to get an index card-like presentation. Furthermore, we implemented a sidebar displaying the course contents in a structured manner, distinguishing the level (i.e., chapter and sub-chapter). Both presentations are adapted from hedonic websites and platforms.

When implementing our new e-learning platform, we focused on providing only enterprise systems-related content. We provided additional courses augmenting the student's enterprise systems end-user training with cross-cutting contents such as process management and selected modeling languages. We completely omitted other contents, such as job-advertisements for companies and other courses not related to enterprise systems, to reduce the potential for distraction on the e-learning platform.

Addressing design principle 6, we implemented support for multiple media formats, such as videos, graphics, text, documents, and audio. We also encouraged course creators to provide a combination of media formats and especially recommend the use of videos in our course creation guidelines. The use of different media formats has a positive effect on the students' satisfaction as they stated that they felt the course would be more interesting this way.



**Fig. 6** We created badges for each chapter. Thereby, we established two different skill-level: trainee and verified. This figure demonstrates an example of the badges for the Procure-to-Pay process



**Fig. 7** Three conversation channels of WLIB: announcements, virtual face-to-face exchange or feedback sessions and group chats. Besides these, there is a support platform and a whatsapp hotline. Students that feel the need for communication might chose which of these channels fits best their comfort

The micro-learning concept has also been adapted while implementing our e-learning platform. This concept has been chosen because flexibility regarding time, location, and content is an important factor for students. Dividing learning content into small heterogeneous and self-contained parts will enable more flexible learning, which in turn leads to a customized learning experience for the students. Furthermore, certain focus areas in which repetition is needed can be quicker detected as the learning units are smaller. We also updated our guidelines for course creators to enable and engage them to transform existing courses to the microlearning<sup>4</sup> concept and create new courses in this format.

Addressing design principle 8, we designed our e-learning platform with progress bars indicating a student's individual progress in each chapter, each sub-chapter, and the whole course. Because of the increased flexibility and the microlearning concept, students might feel lost on how much progress they already made. The progress bar is a helpful feature for trainers as well as for students. In trainer-mode, the trainer can query an overview on the progress of his or her students, the progress they made with the course, and the accomplishments they have achieved.

As the best practice platforms offer gamification elements, we decided to include such elements for the reimplementation of our e-learning platform. These gamification elements should engage students and make learning more fun. Rewards are also a good measure to foster motivation to achieve certain goals. When a student completes a course, there should be a certificate waiting for him or her so that this achievement can be manifested and shared. Also, students participating in one end-user training offered by WLIB claimed that obtaining a certificate would be helpful, as it is proof of their competence. Furthermore, for students to be able to celebrate even small accomplishments during the training

<sup>4</sup> Microlearning is an teaching approach in which learning units are structured in short and focused parts, so that they can be consumed in a short period of time [177, 178]. The microlearning approach might be useful to disentangle the complexity of enterprise systems, to make the concept easier to grasp for students [141]. Furthermore, it has been identified during examining the case study, that most provider apply this concept to their courses. Thus, we aimed at adapting it.

and keeping them motivated, we provided the platform with the possibility of adding badges for each interim accomplishment. We established two skill-level that can be obtained in each chapter: Trainee and Verified. Students obtain the "Trainee-" Badge for accomplishing the basics of each chapter (e.g. Master Data creation, or organizational basics for the respective process that is covered in this chapter). For being able to handle the more complex constructs of each chapter and successfully finishing the test at the end, students obtain the "Verified"-badge. Furthermore, we added the option to apply digital certificates automatically sent when a certain trigger (e.g. finalization of the Training) is met (Fig. 6).

Addressing design principle 10, we implement several channels for inter-student and student-trainer conversation. In WLIB such feedback functionalities are one of the most used features on the e-learning platform. When carrying out tasks within an enterprise system, the students might encounter procedural problems and system errors, which he or she cannot solve by himself or herself. Providing the integration of live support would help, so that the students are able to solve their problems quickly and do not get frustrated. This is particularly important, as the attitude towards the system should be toned positively. We provide a combination of frequently asked questions, help desk, community forum, and weekly virtual consultation hours to speak on certain problems. Trainers can also interact with student through Moodle chat and forums (Fig. 7).

## 6 Evaluation

After the completion of the WLIB implementation and a successful go-live, we were able to evaluate the implemented design principles with students who have successfully finished an enterprise systems end-user training on our e-learning platform. We considered the evaluation successful in case the rating was above average (3). To evaluate our e-learning platform, we created a survey containing parts for evaluating the e-learning platform and selected courses. The meta-requirements gave us direction for formulating the questions regarding the e-learning platform. We opted for using the meta-requirements instead of design principles as, multiple design principles account for a single meta-requirement and thus the design principles are interrelated. To evaluate our proposed meta-requirements and design principles, we created a questionnaire in German language by using the online-tool Limesurvey. As the majority of the learners on our platform (82%) attend the German-based course, we opted for the German language survey. Upon opening the questionnaire, students were welcomed with a short introductory text about the goal of the questionnaire and asked for their consent to participate. When clicking on the "next" button and giving their consent they were asked questions regarding their demographic data. This part was followed by, a section regarding the e-learning platform and a section covering questions about the self-efficacy of the students. The participants were asked to rate different statements according to their perception on a scale from 1 (totally do not agree) to 5 (totally agree). The questionnaire can be found in Appendix E. This questionnaire has been sent to our former participants by email. Of these participants, only 108 gave feedback, which marks a response rate of nearly 9%. Of these 108 respondents, the majority (72.5%) consider themselves male, and they are between 18 to 37 years old (in mean approx. 27 years). 51.9% of participants are students and 44.4% are already employed. Most of the participants (58.7%) are still or have been enrolled at the university of Duisburg-Essen.

The participants rated the design of the learning platform in general in mean with 3.9 (SD = 1.14). Especially the easy accessibility of the platform has been rated positively (M = 4.06, SD = 1.11) and also the flexibility (M = 4.17, SD = 1.149) during course participation (MR1). Furthermore, the participants perceived the structure of the platform as positively (M = 3.79, SD = 1.108) as well as the structure of training (M = 3.74, SD = 1.142) and claimed that the platform was easy to use (M = 3.93; SD = 1.18) (MR2). Furthermore, the motivational elements included in the platform design have been rated positively (M = 3.76; SD = 1.26) (MR3). Seeing these results, we can draw the conclusion that the implementation of our design principles has been successful. Regarding the feasibility of the e-learning platform to enable the exchange between platform users (MR4) the e-learning platform provided by WLIB has been rated positive by participants (M = 3.7, SD = 1.3). The results of the official SAP certification exam confirm the positive evaluation: about 95% of all students on our e-learning platform passed the certification exam.

## 7 Discussion

Our design science research project led to the development and establishment of an e-learning platform focusing on students' end-user training for enterprise systems. A priori to the project, we derived four meta-requirements from a literature review. Addressing these requirements, we developed ten design principles for e-learning platforms focusing

on students' end-user training for enterprise systems. The development of our design principles is informed by an analysis of nine successful e-learning platforms and the implementation of the WLIB platform.

The scientific contribution of this publication is twofold. First, we introduce e-learning platforms as a specific class of information systems that requires explicit design knowledge. We differentiate e-learning platforms among multi-purpose e-learning platforms and end-user training-oriented e-learning platforms. Second, we derive explicit design knowledge in the form of four meta-requirements and a novel set of ten design principles. This design knowledge should provide guidance for the implementation of e-learning platforms tailored for higher-education students' end-user training. This work provides specific guidance for platform owners implementing new and reshaping existing e-learning platforms tailored for enterprise systems end-user training. This research supports the creation of a holistic enterprise system end-user training experience for students. Providing students with enterprise system knowledge is important because the development of enterprise systems usage skills is a lengthy process, and enterprise systems are complex. We further address trainers with a suggested course structure and potential motivational elements that can be integrated into their (new) courses. Furthermore, building upon the design principles provided in this work, guidance on how to design end-user training e-learning platforms for employees in organizations can be adapted and, thus, the end-user training in organizations can be further improved.

In the following, we discuss our design principles in an analytical manner following the framework for minimum reusability evaluation [160, 161]. This framework consists of five criteria (i.e., "accessibility", "importance", "novelty and insightfulness", "actability and guidance" and "effectiveness") and should ensure that design principles provide insights and guidance regarding the creation of IT instances of the same class [84, 160]. Those instances are e-learning platforms, and our design principles can be used by practitioners, such as e-learning platform owners, universities, and companies. Our design principles are accessible because we use language used in the domain. Also, the formulation of our design principles follows the framework designed by Chandra et al. [86]. Our work is important because of the growth in numbers of users in distance learning [28] and the increasing competitive need of companies deploying an enterprise system [162], which again results in the need for end-user training. Using our design principles, we provide a novel approach for designing and implementing an e-learning platform. Furthermore, they can provide an owner with new insights and ideas such as new functionalities for the platform. For each design principle, we suggest different design instances derived from the real world [163], which makes our design principles actable. Our guidance is only prescriptive and platform owners are free to use the design principles for their specific needs. The design principles are effective because they were applicable to the design of our own platform, WLIB.

The failure of many enterprise systems implementation projects can be attributed to the lack of fulfillment of critical success factors [9]. Among others, critical success factors of enterprise systems implementation are proper project management, user training, change management, and end-user acceptance [10]. Most of these critical success factors can be attributed to the end-users of the system and their attitude and knowledge level [13]. Hence, training is an important lever to increase user acceptance and get users to use the system. As enterprise systems are complex systems, end-user training is a complex endeavor, as many different competencies need to be conveyed to the end users. Therefore, traditional training methods often lack efficiency [162]. E-learning can be beneficial to carry out end-user training and might also support the learning process in blended formats. With the use of e-learning, hands-on-sessions can be designed more flexible [12]. The design of e-learning platforms as environments for end-user training require special focus to be efficient [6, 105, 129, 144, 164, 165].

We investigated two types of e-learning platforms in our multi-case analysis. Enterprise systems vendors provide one group, the other group is provided by owners independent from specific enterprise systems. As ownership impacts the content provided via the platform and their design, it is disputable whether enterprise systems vendors or independent owners should provide end-user training e-learning platforms for enterprise systems. Independent owners can allow for multi-homing in the sense, that end-user training for enterprise systems from multiple vendors can be provided [166]. Although we specifically developed our design principles for e-learning platforms with a focus on end-user training for enterprise systems as a class of systems [167], the design principles might be applicable for e-learning platforms with a broader range of courses. We derived design knowledge from seven more general e-learning platforms with some end-user training courses. Hence, the design principles might entail a wider projectability [168].

This research is not without limitations. First, the found design principles are with a focus on enterprise systems end-user training. Nevertheless, there might be information systems that are just as complex as enterprise systems. It might be of interest to investigate, whether these design principles apply for trainings focused on those systems as well. Second, the user interface design has not been focused. As the user attitude plays an important role and the user interface is one factor impacting it, the design in terms of design of the user interface should be of focus when designing e-learning platforms. Third, theories and knowledge about learning processes are only rudimentary taken into consideration. To be able to create an effective learning environment for e-learning-based end-user training for enterprise systems, insights about human cognitive processes should be considered as well. Furthermore, we did not consider the implementation of adaptive design, to support the individualization of learning. Nevertheless, research in this field, especially regarding enterprise systems training, is scarce, and thus, there needs to be a more profound basis in order to be able to give recommendations about this. However, this team of researchers is planning on doing some research in this direction to have a more fruitful basis for discussions in the adaptive learning environment design.

## 8 Conclusion

It has become common practice to integrate enterprise systems end-user trainings in higher-education curricula, to foster students' skills. While e-learning has become a relevant approach in organizational settings to overcome time and spatial constraints in enterprise systems end-user trainings, and several best-practice generalized platform exist, in higher-education settings structured guidance to design these are lacking. To close this gap we developed ten theory-based and empirically validated design principles for the design of e-learning platforms addressing trainers and students of the platform following the approach proposed by Moeller et al. [42]. We focused our design principles on the e-learning platforms for students' end-user training. As a basis for the formation of these design principles served us three data sources: (1) By conducting an extensive literature review, we elicited efficient and flexible learning, easy use, continuous motivation, and exchange between all platform participants (i.e., trainers and students) as meta-requirements for the design of e-learning platforms for students' enterprise systems end-user training. Drawing on these meta-requirements, we derived ten design principles by extending this supportive approach with the (2) analysis of nine cases of successful e-learning platforms. Additionally, we used our gathered design knowledge obtained through (3) the implementation of our own e-learning platform (i.e., WLIB) for students and analyzed the implementation approach and its consequences to update the design principles. This holistic research approach ensures that our ten design principles are applicable to designing e-learning platforms, specifically focusing on end-user training for enterprise systems addressed at higher-education students. For this application scenario, considering its boundaries our proposed design principles are unique [45].

## 9 Outlook

Future research might evaluate our design principles in interviews with domain experts in end-user training and might demonstrate their applicability for general e-learning platforms, and for other application settings [168, 169]. Future research could also provide further design features to improve the implementability of our abstract design principles [90]. These design features can act as additional guideline for implementation. As we intentionally focused our analysis on the e-learning platform matching trainers and students as virtual learning environment, an interesting avenue for future research could be a more detailed investigation of the courses for end-user training. Future research could analyze their didactic concept, course structure, and motivational elements and derive a best practice course concept for enterprise systems end-user training in a virtual setting. Accounting for the different requirements and predispositions of learners, a special focus should be laid on the adaptive design of e-learning platforms Fujs, Vrhovec, and Vavpotič, Lea et al. Future research might also investigate the impact of the application of e-learning platforms on students' computational thinking

[170]. Another interesting avenue for future research would be the impacts of implementing avatar-based learning on e-learning platforms [171]. An avatar-based environment might foster students' communication and collaboration on e-learning platforms [172]. Furthermore, in future research the implementation of new technologies like GenAI to enhance e-learning platforms should be considered, to improve the learning experience, by providing a personalized assistant and thus imitating human communication [173].

**Author contributions** Mareen Wienand Conceptualization-Equal, Methodology-Equal, Validation, Formal Analysis, Resources, Data Curation—Equal, Writing—Original Draft-Equal, Writing—Review and Editing-Equal, Visualization, Supervision-Equal, Project Administration Tobias Wulfert Conceptualization-Equal, Methodology-Equal, Writing—Original Draft-Equal, Writing—Review and Editing-Equal Hiep Hoang Data Curation—Equal, Writing—Original Draft-Equal. The description of the authors' roles is based on the Contributor Roles Taxonomy (CRediT) (<https://credit.niso.org/>).

**Data availability** The Results from the Literature Review (Sec. Supportive), that have been used for derivation of the design principles are available in the appendix Table 1. Appendix Table 2 contains the overview of cases used in the reflective section to deduce further design principles. The analysis of each case is presented in Appendix Tables 3 to 11. The structure of the questionnaire referenced in section 3.3 is presented in Appendix C (Figs. 8, 9). The raw data as well as the detailed results of the evaluation as described in section 5 can be found in the following github repository: <https://github.com/MareenWienand/DesignPrinciplesforElearningPlatforms>. The results indicating the completion rates of the course and the success rates of the SAP certification exams reported in section 5 are available on request. We cannot provide them publicly to preserve individual's privacy under the European General Data Protection Regulation.

## Declarations

**Competing interests** On behalf of all authors, the corresponding author states that there is no competing interest.

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## Appendix A: Concept matrix

In Table 1, we present the literature identified during the systematic literature review in a concept matrix [43].



**Table 1** Concept matrix

| Author                             | Year | Main focus                        |                                     | Meta requirements      |                      |                                    |             |                          |                                         |   |
|------------------------------------|------|-----------------------------------|-------------------------------------|------------------------|----------------------|------------------------------------|-------------|--------------------------|-----------------------------------------|---|
|                                    |      | (e-learning) Plat-<br>form Design | Design of e-learn-<br>ing based EUT | Design of<br>(ERP) EUT | e-learning<br>Design | Effective and<br>flexible learning | Easy to use | Continuous<br>motivation | Exchange<br>between plat-<br>form users |   |
| Rochet and Tirole                  | 2003 | x                                 |                                     |                        |                      |                                    |             |                          |                                         |   |
| Choi, Kim, and Kim                 | 2007 |                                   |                                     | x                      |                      |                                    |             | x                        |                                         |   |
| Santhanam, Sasidharan, and Webster | 2008 |                                   | x                                   |                        |                      |                                    | x           |                          | x                                       |   |
| She et al.                         | 2009 |                                   |                                     |                        | x                    |                                    | x           |                          | x                                       |   |
| Hsu, Yeh, and Yen                  | 2009 | x                                 |                                     |                        |                      |                                    | x           | x                        |                                         |   |
| Gupta, Bostrom, and Huber          | 2010 |                                   | x                                   |                        |                      |                                    |             |                          | x                                       |   |
| Hathorn and Hathorn                | 2010 |                                   |                                     |                        |                      |                                    | x           |                          |                                         | x |
| Tiwana, Konsynski, and Bush        | 2010 | x                                 |                                     |                        |                      |                                    |             |                          |                                         |   |
| Freire, Arezes, and Campos         | 2012 | x                                 |                                     |                        |                      |                                    |             | x                        |                                         |   |
| Kim, Lee, and Ryu                  | 2013 | x                                 |                                     |                        |                      |                                    |             |                          | x                                       | x |
| Ngai, Moon, and Poon               | 2013 |                                   |                                     |                        |                      |                                    | x           |                          |                                         |   |
| Violante and Vezzetti              | 2014 | x                                 |                                     |                        |                      |                                    | x           |                          |                                         | x |
| Teraes and Herrington              | 2014 |                                   |                                     |                        |                      |                                    | x           |                          |                                         | x |
| Benta, Bologna, and Dzitac         | 2014 | x                                 |                                     |                        |                      |                                    |             |                          |                                         | x |
| Hagiu and Wright                   | 2015 | x                                 |                                     |                        |                      |                                    | x           |                          |                                         | x |
| Milligan and Griffin               | 2016 |                                   |                                     |                        |                      |                                    | x           |                          |                                         |   |
| Liu                                | 2017 | x                                 |                                     |                        |                      |                                    |             |                          | x                                       | x |
| Schwade and Schubert               | 2018 | x                                 |                                     |                        |                      | x                                  |             |                          | x                                       |   |
| Asadullah, Faik, and Kankanhalli   | 2018 | x                                 |                                     |                        |                      |                                    |             |                          |                                         |   |
| Cacheiro-Gonzalez et al.           | 2019 | x                                 |                                     |                        |                      |                                    | x           |                          |                                         | x |
| Haj-Bolouri                        | 2019 |                                   |                                     | x                      |                      |                                    |             |                          |                                         |   |
| Nah et al.                         | 2019 |                                   |                                     |                        |                      |                                    |             |                          |                                         |   |
| Pridmore and Godin                 | 2020 |                                   |                                     |                        |                      |                                    | x           |                          | x                                       |   |
| Nordin, Singh, and Mansor          | 2020 |                                   |                                     |                        |                      | x                                  | x           |                          |                                         |   |
| Donath, Mircea, and Rozman         | 2020 | x                                 |                                     |                        |                      |                                    | x           |                          |                                         |   |
| Javaid et al.                      | 2021 |                                   |                                     |                        |                      | x                                  | x           |                          | x                                       | x |
| McCarthy et al.                    | 2021 | x                                 |                                     |                        |                      |                                    | x           |                          | x                                       |   |
| Nordin, Singh, and Mansor          | 2021 |                                   |                                     |                        |                      | x                                  | x           |                          |                                         |   |
| Li                                 | 2021 | x                                 |                                     |                        |                      |                                    | x           |                          |                                         | x |
| Soellner                           | 2021 |                                   | x                                   |                        |                      |                                    | x           |                          |                                         |   |
| Gawer                              | 2021 | x                                 |                                     |                        |                      |                                    |             |                          |                                         |   |
| Zhang                              | 2022 |                                   |                                     |                        |                      |                                    |             |                          | x                                       | x |
| Lea et al.                         | 2022 |                                   |                                     |                        |                      |                                    | x           |                          |                                         |   |
| Suemer and Aydin                   | 2022 |                                   |                                     |                        |                      |                                    |             |                          | x                                       | x |

Table 1 (continued)

| Author                                              | Year | Main focus<br>(e-learning) Plat-<br>form Design | Meta requirements                   |                        |                      |                                    |             |                          |
|-----------------------------------------------------|------|-------------------------------------------------|-------------------------------------|------------------------|----------------------|------------------------------------|-------------|--------------------------|
|                                                     |      |                                                 | Design of e-learn-<br>ing based EUT | Design of<br>(ERP) EUT | e-learning<br>Design | Effective and<br>flexible learning | Easy to use | Continuous<br>motivation |
| Zhang and Locke<br>Qian<br>Vlachogianni and Tselios | 2022 | x                                               |                                     |                        |                      | x                                  | x           |                          |
|                                                     | 2022 | x                                               |                                     |                        |                      | x                                  | x           |                          |
|                                                     | 2022 | x                                               |                                     |                        |                      | x                                  | x           |                          |

## Appendix B: Overview of cases

Table 2 shows all cases, we considered in our case analysis.

**Table 2** Overview of cases

| Company platform       | Est. | No. users    | No. courses | No. lecturers | Revenue (USD) | Target group                    | Topics     |
|------------------------|------|--------------|-------------|---------------|---------------|---------------------------------|------------|
| OpenSAP [179]          | 2013 | 1.1 million  | 250         | n.a.          | 30 billion    | Business                        | SAP        |
| Udemy                  | 2010 | 57 million   | 213.000     | 74.000        | 518.2 million | Business/private                | General    |
| EdX                    | 2012 | 42 million   | 3.600       | 15.000        | n.a.          | Business/private                | General    |
| Coursera               | 2012 | 113 million  | 5.400       | n.a.          | 415.3 million | Business/private                | General    |
| Udacity                | 2012 | 16.9 million | n.a.        | n.a.          | 414.9 million | Business/private/<br>government | General    |
| Iversity               | 2013 | n.a.         | n.a.        | n.a.          | 1.7 million   | Business/private                | General    |
| Skillshare             | 2011 | 800.000      | 35.000      | 11.000        | n.a.          | Business/private                | General    |
| FutureLearn            | 2013 | n.a.         | 1.000       | n.a.          | 13.5 million  | Business/private                | General    |
| Trailhead (Salesforce) | 2014 | 4 million    | n.a.        | n.a.          | 26.5 billion  | Business                        | Salesforce |

## Appendix C: Case information

In this section, we present exemplary websites from which we extracted information for the derivation of the design principles (Tables 3, 4, 5, 6, 7, 8, 9, 10, 11).

**Table 3** Exemplary websites analyzed for coursera

| Design principle                        | URL                                                                     | Description                                                   |
|-----------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------|
| Accessibility                           | <a href="https://tinyurl.com/bdfpwe9y">https://tinyurl.com/bdfpwe9y</a> | Login screen                                                  |
| Accessibility                           | <a href="https://tinyurl.com/4t525zp2">https://tinyurl.com/4t525zp2</a> | Start of a module/course                                      |
| Permanent provision of learning content | <a href="https://tinyurl.com/37rcppvt">https://tinyurl.com/37rcppvt</a> | Overview of the learning content and display of the progress  |
| Interactive tasks                       | <a href="https://tinyurl.com/37rcppvt">https://tinyurl.com/37rcppvt</a> | Conduct quizzes                                               |
| Interactive tasks                       | <a href="https://tinyurl.com/5etvhf8u">https://tinyurl.com/5etvhf8u</a> | Conduct quizzes                                               |
| Clear structure and simple navigation   | <a href="https://tinyurl.com/355akwps">https://tinyurl.com/355akwps</a> | Pre-filtered selection of course content and search function  |
| Only course-related aspects             | <a href="https://tinyurl.com/v4rnd9pb">https://tinyurl.com/v4rnd9pb</a> | Overview of the topics of the courses                         |
| Multimedia                              | <a href="https://tinyurl.com/2bh83ky4">https://tinyurl.com/2bh83ky4</a> | Large variety of media                                        |
| Multimedia                              | <a href="https://tinyurl.com/ypctujdz">https://tinyurl.com/ypctujdz</a> | Large variety of media                                        |
| Didactical elements                     | <a href="https://tinyurl.com/rhsz5e34">https://tinyurl.com/rhsz5e34</a> | Division of the course into many submodules                   |
| Didactical elements                     | <a href="https://tinyurl.com/2p9m4mdh">https://tinyurl.com/2p9m4mdh</a> | Division of the course into many submodules                   |
| Track of course progress                | <a href="https://tinyurl.com/3nncb222">https://tinyurl.com/3nncb222</a> | Detailed course overview                                      |
| Motivational elements                   | <a href="https://tinyurl.com/ykukxpmg">https://tinyurl.com/ykukxpmg</a> | Achieving a degree                                            |
| Motivational elements                   | <a href="https://tinyurl.com/yvkd3d4s">https://tinyurl.com/yvkd3d4s</a> | Certificate acquisition                                       |
| Communication                           | <a href="https://tinyurl.com/bdh6tbd4">https://tinyurl.com/bdh6tbd4</a> | Possibility to get in touch with a teacher, e.g. via LinkedIn |
| Communication                           | <a href="https://tinyurl.com/5n8uahzh">https://tinyurl.com/5n8uahzh</a> | Possibility to look at the evaluations of the course          |

**Table 4** Exemplary websites analyzed for edx

| Design principle                        | URL                                                                     | Description                                                  |
|-----------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------|
| Accessibility                           | <a href="https://tinyurl.com/49n8zcp2">https://tinyurl.com/49n8zcp2</a> | Login screen                                                 |
| Accessibility                           | <a href="https://tinyurl.com/337vv936">https://tinyurl.com/337vv936</a> | Start of a module/course                                     |
| Permanent provision of learning content | <a href="https://tinyurl.com/mr3e8j5n">https://tinyurl.com/mr3e8j5n</a> | Overview of the learning content and display of the progress |
| Interactive tasks                       | <a href="https://tinyurl.com/mr3e8j5n">https://tinyurl.com/mr3e8j5n</a> | Conduct quizzes                                              |
| Clear structure and simple navigation   | <a href="https://tinyurl.com/46pmchdk">https://tinyurl.com/46pmchdk</a> | Pre-filtered selection of course content and search function |
| Only course-related aspects             | <a href="https://tinyurl.com/yc8u4jmp">https://tinyurl.com/yc8u4jmp</a> | Overview of the topics of the courses                        |
| Multimedia                              | <a href="https://tinyurl.com/2c2xsbed">https://tinyurl.com/2c2xsbed</a> | Large variety of media                                       |
| Multimedia                              | <a href="https://tinyurl.com/352294px">https://tinyurl.com/352294px</a> | Large variety of media                                       |
| Didactical elements                     | <a href="https://tinyurl.com/c6bv7jnf">https://tinyurl.com/c6bv7jnf</a> | Division of the course into many submodules                  |
| Didactical elements                     | <a href="https://tinyurl.com/2ey9fker">https://tinyurl.com/2ey9fker</a> | Division of the course into many submodules                  |
| Track of course progress                | <a href="https://tinyurl.com/c6bv7jnf">https://tinyurl.com/c6bv7jnf</a> | Detailed course overview                                     |
| Motivational elements                   | <a href="https://tinyurl.com/mr3b462z">https://tinyurl.com/mr3b462z</a> | Certificate acquisition                                      |
| Communication                           | <a href="https://tinyurl.com/5n8ynckj">https://tinyurl.com/5n8ynckj</a> | Discussion forum                                             |

**Table 5** Exemplary websites analyzed for FutureLearn

| Design Principle                        | URL                                                                     | Description                                                  |
|-----------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------|
| Accessibility                           | <a href="https://tinyurl.com/28wb8x44">https://tinyurl.com/28wb8x44</a> | Login screen                                                 |
| Accessibility                           | <a href="https://tinyurl.com/9tbpavms">https://tinyurl.com/9tbpavms</a> | Start of a module/course                                     |
| Permanent provision of learning content | <a href="https://tinyurl.com/yxmbnhdm">https://tinyurl.com/yxmbnhdm</a> | Overview of the learning content and display of the progress |
| Interactive tasks                       | <a href="https://tinyurl.com/mvuzdp38">https://tinyurl.com/mvuzdp38</a> | Conduct quizzes                                              |
| Clear structure and simple navigation   | <a href="https://www.futurelearn.com">https://www.futurelearn.com</a>   | Pre-filtered selection of course content and search function |
| Only course-related aspects             | <a href="https://tinyurl.com/49sy5yew">https://tinyurl.com/49sy5yew</a> | Overview of the topics of the courses                        |
| Multimedia                              | <a href="https://tinyurl.com/mvuzdp38">https://tinyurl.com/mvuzdp38</a> | Large variety of media                                       |
| Didactical elements                     | <a href="https://tinyurl.com/4e7jbpuu">https://tinyurl.com/4e7jbpuu</a> | Division of the course into many submodules                  |
| Didactical elements                     | <a href="https://tinyurl.com/2tdpj3pa">https://tinyurl.com/2tdpj3pa</a> | Division of the course into many submodules                  |
| Track of course progress                | <a href="https://tinyurl.com/mvuzdp38">https://tinyurl.com/mvuzdp38</a> | Detailed course overview                                     |
| Motivational elements                   | <a href="https://tinyurl.com/yshfkwn">https://tinyurl.com/yshfkwn</a>   | Certificate acquisition                                      |
| Motivational elements                   | <a href="https://tinyurl.com/px6wshxv">https://tinyurl.com/px6wshxv</a> | Certificate acquisition                                      |
| Communication                           | <a href="https://tinyurl.com/46zpfcb">https://tinyurl.com/46zpfcb</a>   | Ability to join a virtual community to learn together        |

**Table 6** Exemplary websites analyzed for iversity

| Design principle                        | URL                                                                     | Description                                                  |
|-----------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------|
| Accessibility                           | <a href="https://tinyurl.com/ywk72vvh">https://tinyurl.com/ywk72vvh</a> | Login screen                                                 |
| Accessibility                           | <a href="https://tinyurl.com/2y7kj45m">https://tinyurl.com/2y7kj45m</a> | Start of a module/course                                     |
| Permanent provision of learning content | <a href="https://tinyurl.com/bduhhp9d">https://tinyurl.com/bduhhp9d</a> | Overview of the learning content and display of the progress |
| Interactive tasks                       | <a href="https://tinyurl.com/5749w252">https://tinyurl.com/5749w252</a> | Conduct quizzes                                              |
| Clear structure and simple navigation   | <a href="https://tinyurl.com/y4f53xk3">https://tinyurl.com/y4f53xk3</a> | Pre-filtered selection of course content and search function |
| Only course-related aspects             | <a href="https://tinyurl.com/y8whrhcc">https://tinyurl.com/y8whrhcc</a> | Overview of the topics of the courses                        |
| Multimedia                              | <a href="https://tinyurl.com/5749w252">https://tinyurl.com/5749w252</a> | Large variety of media                                       |
| Didactical elements                     | <a href="https://tinyurl.com/y8whrhcc">https://tinyurl.com/y8whrhcc</a> | Division of the course into many submodules                  |
| Track of course progress                | <a href="https://tinyurl.com/y8whrhcc">https://tinyurl.com/y8whrhcc</a> | Detailed course overview                                     |
| Motivational elements                   | <a href="https://tinyurl.com/5749w252">https://tinyurl.com/5749w252</a> | Certificate acquisition                                      |
| Communication                           | <a href="https://tinyurl.com/39atnn25">https://tinyurl.com/39atnn25</a> | Discussion forum                                             |

**Table 7** Exemplary websites analyzed for OpenSAP

| Design principle                        | URL                                                                     | Description                                                  |
|-----------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------|
| Accessibility                           | <a href="https://tinyurl.com/5yhhcsa6">https://tinyurl.com/5yhhcsa6</a> | Login screen                                                 |
| Accessibility                           | <a href="https://tinyurl.com/y75ant79">https://tinyurl.com/y75ant79</a> | Start of a module/course                                     |
| Permanent provision of learning content | <a href="https://tinyurl.com/3bkkaneb">https://tinyurl.com/3bkkaneb</a> | Overview of the learning content and display of the progress |
| Interactive tasks                       | <a href="https://tinyurl.com/bd8ddmh9">https://tinyurl.com/bd8ddmh9</a> | Conduct quizzes                                              |
| Clear structure and simple navigation   | <a href="https://tinyurl.com/mtx5w4">https://tinyurl.com/mtx5w4</a>     | Pre-filtered selection of course content and search function |
| Only course-related aspects             | <a href="https://tinyurl.com/mtx5w4">https://tinyurl.com/mtx5w4</a>     | Overview of the topics of the courses                        |
| Multimedia                              | <a href="https://tinyurl.com/y75ant79">https://tinyurl.com/y75ant79</a> | Large variety of media                                       |
| Didactical elements                     | <a href="https://tinyurl.com/27296s4x">https://tinyurl.com/27296s4x</a> | Division of the course into many submodules                  |
| Didactical elements                     | <a href="https://tinyurl.com/mvc2psyc">https://tinyurl.com/mvc2psyc</a> | Division of the course into many submodules                  |
| Track of course progress                | <a href="https://tinyurl.com/y75ant79">https://tinyurl.com/y75ant79</a> | Detailed course overview                                     |
| Motivational elements                   | <a href="https://tinyurl.com/y75ant79">https://tinyurl.com/y75ant79</a> | Certificate acquisition                                      |
| Motivational elements                   | <a href="https://tinyurl.com/58j7ahds">https://tinyurl.com/58j7ahds</a> | Certificate acquisition                                      |
| Communication                           | <a href="https://tinyurl.com/yckyfv8j">https://tinyurl.com/yckyfv8j</a> | Discussion forum                                             |
| Communication                           | <a href="https://tinyurl.com/5cw58w22">https://tinyurl.com/5cw58w22</a> | Possibility to join a learning room                          |

**Table 8** Exemplary websites analyzed for SkillShare

| Design principle                        | URL                                                                     | Description                                                  |
|-----------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------|
| Accessibility                           | <a href="https://tinyurl.com/yc5r8juk">https://tinyurl.com/yc5r8juk</a> | Login screen                                                 |
| Accessibility                           | <a href="https://tinyurl.com/6d5hh375">https://tinyurl.com/6d5hh375</a> | Start of a module/course                                     |
| Permanent provision of learning content | <a href="https://tinyurl.com/6d5hh375">https://tinyurl.com/6d5hh375</a> | Overview of the learning content and display of the progress |
| Interactive tasks                       | <a href="https://tinyurl.com/6d5hh375">https://tinyurl.com/6d5hh375</a> | Conduct quizzes                                              |
| Clear structure and simple navigation   | <a href="https://tinyurl.com/ycya6n62">https://tinyurl.com/ycya6n62</a> | Pre-filtered selection of course content and search function |
| Only course-related aspects             | <a href="https://tinyurl.com/6d5hh375">https://tinyurl.com/6d5hh375</a> | Overview of the topics of the courses                        |
| Multimedia                              | <a href="https://tinyurl.com/6d5hh375">https://tinyurl.com/6d5hh375</a> | Large variety of media                                       |
| Didactical elements                     | <a href="https://tinyurl.com/6d5hh375">https://tinyurl.com/6d5hh375</a> | Division of the course into many submodules                  |
| Track of course progress                | <a href="https://tinyurl.com/6d5hh375">https://tinyurl.com/6d5hh375</a> | Detailed course overview                                     |
| Motivational elements                   | <a href="https://tinyurl.com/ypz3366z">https://tinyurl.com/ypz3366z</a> | Certificate acquisition                                      |
| Communication                           | <a href="https://tinyurl.com/6d5hh375">https://tinyurl.com/6d5hh375</a> | Possibility to look at the evaluations of the course         |

**Table 9** Exemplary websites analyzed for trailhead

| Design principle                        | URL                                                                     | Description                                                  |
|-----------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------|
| Accessibility                           | <a href="https://tinyurl.com/2ytprhsz">https://tinyurl.com/2ytprhsz</a> | Login screen                                                 |
| Accessibility                           | <a href="https://tinyurl.com/2bevbnyr">https://tinyurl.com/2bevbnyr</a> | Start of a module/course                                     |
| Permanent provision of learning content | <a href="https://tinyurl.com/34rhahxh">https://tinyurl.com/34rhahxh</a> | Overview of the learning content and display of the progress |
| Interactive tasks                       | <a href="https://tinyurl.com/mrxk93re">https://tinyurl.com/mrxk93re</a> | Conduct quizzes                                              |
| Clear structure and simple navigation   | <a href="https://tinyurl.com/yvfuaehk">https://tinyurl.com/yvfuaehk</a> | Pre-filtered selection of course content and search function |
| Only course-related aspects             | <a href="https://tinyurl.com/mry257xk">https://tinyurl.com/mry257xk</a> | Overview of the topics of the courses                        |
| Multimedia                              | <a href="https://tinyurl.com/mrxk93re">https://tinyurl.com/mrxk93re</a> | Large variety of media                                       |
| Didactical elements                     | <a href="https://tinyurl.com/3jkdfaxp">https://tinyurl.com/3jkdfaxp</a> | Division of the course into many submodules                  |
| Didactical elements                     | <a href="https://tinyurl.com/53fhzpkf">https://tinyurl.com/53fhzpkf</a> | Division of the course into many submodules                  |
| Track of course progress                | <a href="https://tinyurl.com/yppf5myd">https://tinyurl.com/yppf5myd</a> | Detailed course overview                                     |
| Motivational elements                   | <a href="https://tinyurl.com/y3c7ufdn">https://tinyurl.com/y3c7ufdn</a> | Superbadge acquisition                                       |
| Motivational elements                   | <a href="https://tinyurl.com/4ezp88n3">https://tinyurl.com/4ezp88n3</a> | Certificate acquisition                                      |
| Communication                           | <a href="https://tinyurl.com/5f5nuswz">https://tinyurl.com/5f5nuswz</a> | Discussion within a group                                    |



**Table 10** Exemplary websites analyzed for udacity

| Design principle                        | URL                                                                     | Description                                                  |
|-----------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------|
| Accessibility                           | <a href="https://tinyurl.com/48xf68xu">https://tinyurl.com/48xf68xu</a> | Login screen                                                 |
| Accessibility                           | <a href="https://tinyurl.com/y7zznjpf">https://tinyurl.com/y7zznjpf</a> | Start of a module/course                                     |
| Permanent provision of learning content | <a href="https://tinyurl.com/y7zznjpf">https://tinyurl.com/y7zznjpf</a> | Overview of the learning content and display of the progress |
| Interactive tasks                       | <a href="https://tinyurl.com/4hywmwuf">https://tinyurl.com/4hywmwuf</a> | Conduct quizzes                                              |
| Clear structure and simple navigation   | <a href="https://tinyurl.com/mrxh9zhy">https://tinyurl.com/mrxh9zhy</a> | Pre-filtered selection of course content and search function |
| Only course-related aspects             | <a href="https://tinyurl.com/mtpv7dva">https://tinyurl.com/mtpv7dva</a> | Overview of the topics of the courses                        |
| Multimedia                              | <a href="https://tinyurl.com/4vtazuen">https://tinyurl.com/4vtazuen</a> | Large variety of media                                       |
| Didactical elements                     | <a href="https://tinyurl.com/3pxt66j8">https://tinyurl.com/3pxt66j8</a> | Division of the course into many submodules                  |
| Didactical elements                     | <a href="https://tinyurl.com/2s3bcnu4">https://tinyurl.com/2s3bcnu4</a> | Division of the course into many submodules                  |
| Track of course progress                | <a href="https://tinyurl.com/2s3bcnu4">https://tinyurl.com/2s3bcnu4</a> | Detailed course overview                                     |
| Motivational elements                   | <a href="https://tinyurl.com/mwtkfjen">https://tinyurl.com/mwtkfjen</a> | Certificate acquisition                                      |
| Communication                           | <a href="https://tinyurl.com/mwtkfjen">https://tinyurl.com/mwtkfjen</a> | Possibility to look at the evaluations of the course         |

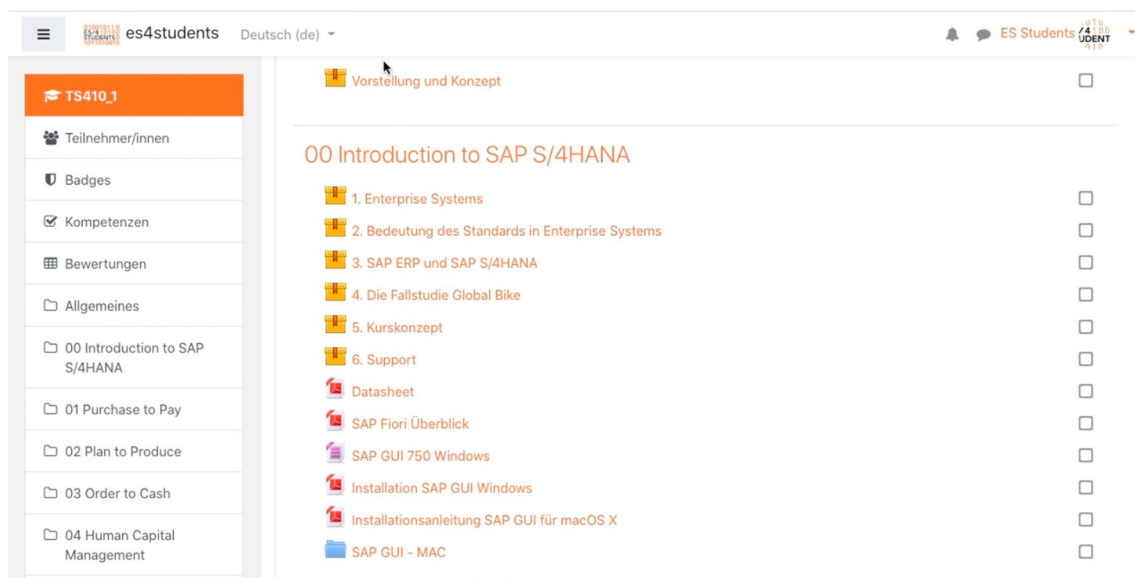
**Table 11** Exemplary websites analyzed for udemy

| Design principle                        | URL                                                                     | Description                                                  |
|-----------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------|
| Accessibility                           | <a href="https://tinyurl.com/3djcwpaH">https://tinyurl.com/3djcwpaH</a> | Login screen                                                 |
| Accessibility                           | <a href="https://tinyurl.com/2d2387ry">https://tinyurl.com/2d2387ry</a> | Start of a module/course                                     |
| Permanent provision of learning content | <a href="https://tinyurl.com/2h8wfzFz">https://tinyurl.com/2h8wfzFz</a> | Overview of the learning content and display of the progress |
| Interactive tasks                       | <a href="https://tinyurl.com/yhatad3v">https://tinyurl.com/yhatad3v</a> | Conduct quizzes                                              |
| Clear structure and simple navigation   | <a href="https://tinyurl.com/5n7b4rnX">https://tinyurl.com/5n7b4rnX</a> | Pre-filtered selection of course content and search function |
| Only course-related aspects             | <a href="https://tinyurl.com/5y3chs46">https://tinyurl.com/5y3chs46</a> | Overview of the topics of the courses                        |
| Multimedia                              | <a href="https://tinyurl.com/yhatad3v">https://tinyurl.com/yhatad3v</a> | Large variety of media                                       |
| Didactical elements                     | <a href="https://tinyurl.com/5y3chs46">https://tinyurl.com/5y3chs46</a> | Division of the course into many sub-modules                 |
| Didactical elements                     | <a href="https://tinyurl.com/2h8wfzFz">https://tinyurl.com/2h8wfzFz</a> | Division of the course into many sub-modules                 |
| Track of course progress                | <a href="https://tinyurl.com/5y3chs46">https://tinyurl.com/5y3chs46</a> | Detailed course overview                                     |
| Motivational elements                   | <a href="https://tinyurl.com/4djevaw9">https://tinyurl.com/4djevaw9</a> | Certificate acquisition                                      |
| Communication                           | <a href="https://tinyurl.com/4djevaw9">https://tinyurl.com/4djevaw9</a> | Possibility of Q & A and direct messages with teacher        |

## Appendix D: Iterative design

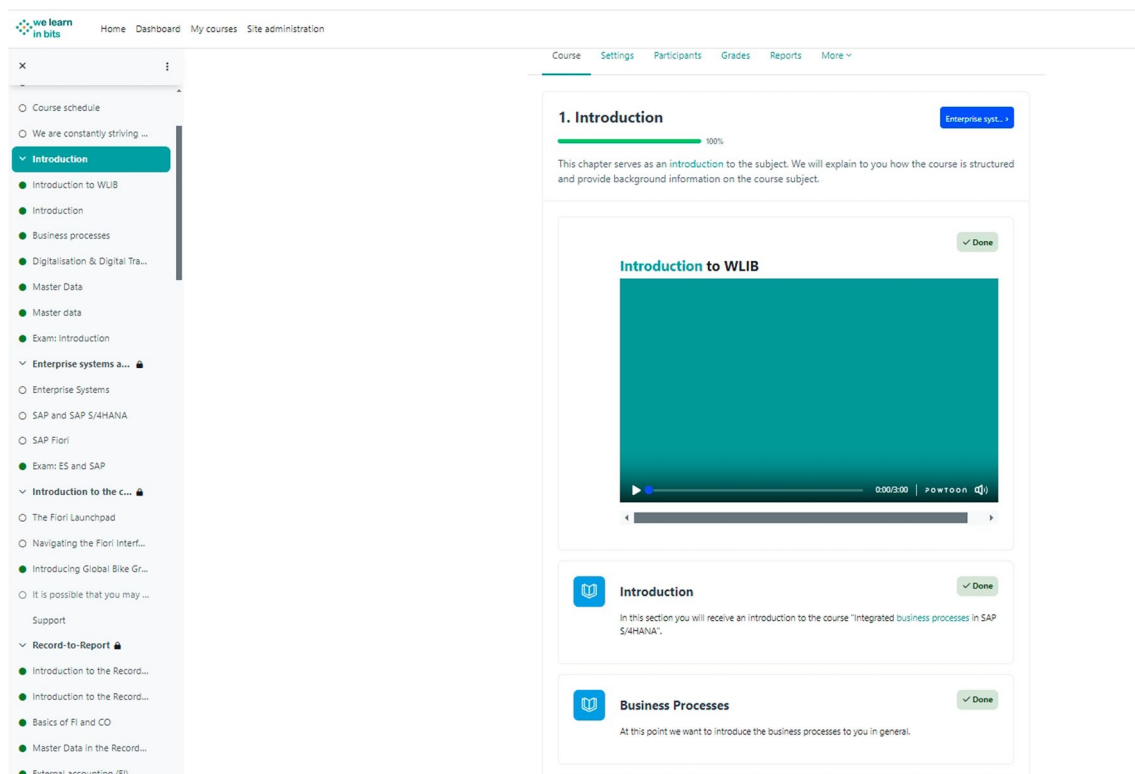
We started our iterative design with a basic Moodle template (see Fig. 8). In the first step, we changed the appearance of the platform by using custom templates to make it more aesthetically comforting and to increase its usability (e.g., by adding tiles for navigation). Furthermore, we changed the colors of the platform, as there is evidence, that colors hues of red, like orange that is used in the initial template of Moodle, have a negative impact on users, whilst others, especially the hues of blue have a rather positive effect. Furthermore, a light environment is perceived as engaging [174, 175].

In the first step, we changed the appearance of the platform by using custom templates to make it more aesthetically comforting and to increase its usability (e.g., by adding tiles for navigation). Furthermore, we changed the colors of the platform, as there is evidence, that colors hues of red, like orange that is used in the initial template of Moodle, have a negative impact on users, whilst others, especially the hues of blue have a rather positive effect. Furthermore, a light environment is perceived as engaging [174, 175]. We initially provided information about job offerings or other information about organizations on our platform for the students. This information seemed to distract and thus, we omitted everything that was not related to the course content from the platform, to enable efficient and flexible learning. Successively we added elements like integrated support platforms, an integrated shop, single-sign-on options, and an improved navigation bar improve handling of the platform and support easy use. Furthermore, we provided interactive elements like quizzes, or progress tracking to keep the students motivated during use of the platform. Finally, we increased the possibility of exchanging information between platform users



**Fig. 8** The basis of our considerations of implementing WLIB. The design was a basic Moodle design and the clarity of navigation and structure of page was missing

and trainers, as one major critique point that has been named by former users was a lack of communication. We adapted the platform iteratively over the course of 4 years, before we were able to finally implement WLIB platform in 2021. Since then smaller iterative circles have been undertaken, that were followed by only minor changes to the platform design. (See Fig. 9). Since the re-implementation of our platform, we have been able to increase the number



**Fig. 9** The design of our final WLIB platform. Compared to the starting point the design was chosen more appealing and with a more defined structure

of courses offered from initially three to 14, as the platform provides benefits not only to the learner (which is the focus of this publication) but also to the lecturers. Since then we have received more than 670 orders and sold more than 800 courses, More than 750 participants<sup>5</sup> were able to complete the courses in this period successfully. About 300 of those participants chose a course with a certification option, and only 12 of those participants failed the exam, which led to a success rate of 96.1 %.

## Appendix E: Structure of questionnaire

Supplementary material presents the structure of the questionnaire that has been used for evaluation.

## Appendix F: Raw data

The raw data of the evaluation survey is publicly available at: [GitHub:Results Evaluation](#).

## Appendix G: Results

The results of the evaluation survey are publicly available at [GitHub:Results Evaluation](#).

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<sup>5</sup> The number of participants differs from the number of orders as there 6 group enrolments comprising 15–26 participants each.

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